

Rory McCulloch

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Seeking Summer 2027 international co-op opportunities in AI/ML and embedded systems.

EXPERIENCE

Trench Group – Project NOVA

May 2026 – Oct 2026

AI/ML Applied Sciences Intern

- **Built an internal SAP sales-quoting tool running on rule-based logic, Volt**, and led a re-architecture that moved model-selection and configuration logic into a companion tool, Spark, that extracts structured data from customer RFQs via **an LLM/NLP pipeline**. (Azure OpenAI, spaCy)
- Diagnosed and fixed a 3-stage latency bottleneck in Volt via connection pooling, query caching, and a database region migration (Canada East → West Europe), cut request latency **89% (4.2s → 450ms)** and downstream quote-generation latency **96% (18s → ~750ms)**. (Python, pyodbc, Azure SQL)
- Extended Volt from single-product (CVT) to **multi-product automated quoting (CVT, TPVT, CC)**, generating draft SAP quotations for Trench's global technical sales team.

Aim Colours

May 2025 – Sep 2025

Electrical Engineering Intern

- Designed and routed **multi-layer PCB backplanes** for wearable technology, partnering with industrial designers to meet strict ergonomic form-factor constraints. (KiCad, Altium, Embedded Programming)
- Analyzed power consumption profiles for embedded microcontrollers (XIAO MG24) and streamlined DFM validation and hardware-verification workflows, **shrinking overall component footprint by 2x**.

PROJECTS

Hands-Up – Gesture Recognition on Embedded Hardware

Jul 2026

Personal Project

- Built an end-to-end embedded ML pipeline: an STM32 Nucleo streams an 8x8 depth grid from a VL53L8CX time-of-flight sensor over serial to a **PyTorch classifier** for real-time rock/paper/scissors gesture recognition.
- Diagnosed a class-imbalance failure, rebuilt the data-collection workflow, and added best-epoch checkpointing to reach **80.8%** validation accuracy and **78.9%** on a true held-out test set. (Python, PyTorch, C++)

Shazam4Drones – Defense Tech Hackathon

Nov 2025

Counter-UAV System

- Built a real-time drone detection system using audio classification, reaching **70% detection accuracy**; placed **4th of 87 teams** at a national defense-tech hackathon. (Python, PyTorch)
- Engineered a deployable counter-UAV pipeline integrating audio classification with a real-time signal-processing backend, and adapted a battlefield mobile interface into a standalone embedded device for field use.

LEADERSHIP

IEEE uOttawa Student Branch

Sep 2024 – Present

Vice Chair, Formerly VP External

- Secured **30+ industry sponsors and partners** across IEEE and the Student Professional Awareness Conference (SPAC, **300+ attendees**), leading outreach strategy and managing two Industry Coordinators.
- Organized a Ciena Networking Night (**200+ students**, 20 engineers) and a Resume Roast panel with interns from Microsoft, Tesla, Meta, AMD, Nokia, and P&G.

EDUCATION

University of Ottawa

Sep 2024 – Apr 2029

BASc in Electrical Engineering & BSc in Computing Technology (Dual Degree)

Relevant coursework: Computer Architecture, Circuit Theory I & II, Data Structures and Algorithms

SKILLS

Languages: Python, C++, Java, Go, R, JavaScript

AI/ML & Data: PyTorch, scikit-learn, NumPy, pandas

Hardware & Tools: KiCad, Arduino, Visual Studio Code, Git